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A study of the relationship between hydrocarbon migration and the precipitation of authigenic magnetic minerals in the Triassic Chugwater Formation, southern Montana

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ABSTRACT

To test the hypothesis that hydrocarbons can cause the precipitation of authigenic magnetite, hydrocarbon-saturated, bleached, and unaltered samples of the Chugwater red beds were examined, using petrographic, rock-magnetic, and paleomagnetic techniques. Samples were collected from the crest of Red Dome in Carbon County, Montana, where the Chugwater is saturated with hydrocarbons and bleached around fractures. The fractures formed during Laramide deformation and provided conduits for oil migration.

The red beds contain Triassic and rotated Triassic magnetizations, which reside in hematite. The magnetization of the oil-saturated sandstones predominantly resides in magnetite, and there is some rock-magnetic evidence for pyrrhotite. The intensity of the natural remanent magnetization (NRM) in these sands is at least an order of magnitude less than that observed in the red beds. A stable magnetic direction could not be isolated from the oil-saturated sandstones. Petrographic studies indicate that both detrital and authigenic magnetite are present in the oil-saturated sandstones. The bleached specimens possess intermediate magnetic characteristics and contain abundant secondary authigenic specular hematite. These specimens contain an early Tertiary magnetization that resides in the authigenic hematite.

The chemical conditions created by the hydrocarbons caused the dissolution of the hematite, which "uncovered" and relatively enhanced the magnetization residing in detrital magnetite. It is also likely that hydrocarbons caused the precipitation of some authigenic magnetite and pyrrhotite. The early Tertiary magnetization in the bleached sandstones is interpreted to be of chemical origin and a byproduct of hydrocarbon migration at Red Dome. The magnetization was probably acquired when the dissolved iron from the red beds precipitated as specular hematite. The results of this study have implications for understanding hydrocarbon-induced diagenesis and remagnetization mechanisms, and for aeromagnetic hydrocarbon exploration.

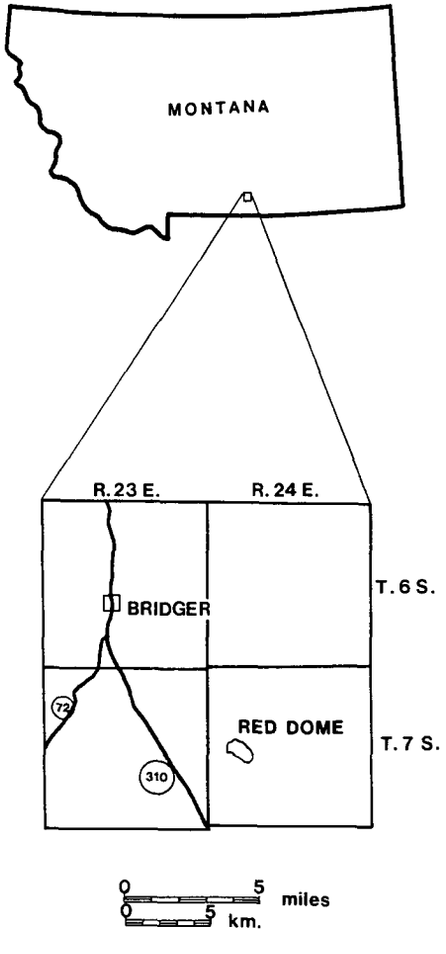


Figure 1. Map of study area, showing the location of Red Dome in south-central Montana.

INTRODUCTION

Recent work has established that there is a relationship between hydrocarbons and precipitation of authigenic magnetic minerals (Elmore and others, 1987; McCabe and others, 1987). Establishment of the relationship has important implications for hydrocarbon exploration, understanding remagnetization mechanisms, and development of a paleomagnetic approach to date hydrocarbon migration. Although the evidence indicates that hydrocarbons can cause precipitation of authigenic magnetite in limestones (for example, Elmore and others, 1987), additional work is needed to test for a relationship in other lithologies, such as sandstones.

In this paper, we report results from a study of the Triassic Chugwater Formation in which we tested the hypothesis that hydrocarbons cause precipitation of authigenic magnetic minerals in a sandstone unit. The test location is at the crest of Red Dome in southern Montana, where the Chugwater red beds are bleached white along either side of hydrocarbon-filled fractures (Figs. 1 and 2). Samples of the hydrocarbon-saturated, bleached, and unaltered sandstones were collected and analyzed using paleomagnetic, rock-magnetic, and petrographic techniques.

The geologic conditions at the study location are appropriate for a test of the hydrocarbon-magnetite hypothesis for several reasons. First, the rock-magnetic and paleomagnetic properties of the Chugwater red beds are well documented (for example, Collinson, 1965, 1967, 1968a, 1968b, 1974), and hematite carrying a Triassic magnetization is the predominant magnetic mineral in the formation (for example, Shive and others, 1984). Only minor quantities of detrital magnetite are found in the formation and are not considered to be significant contributors to the natural remanent magnetization (Shive and others, 1984). Therefore, if abundant magnetite is found in the bleached zones or in the oil-impregnated sandstones, it is likely to be

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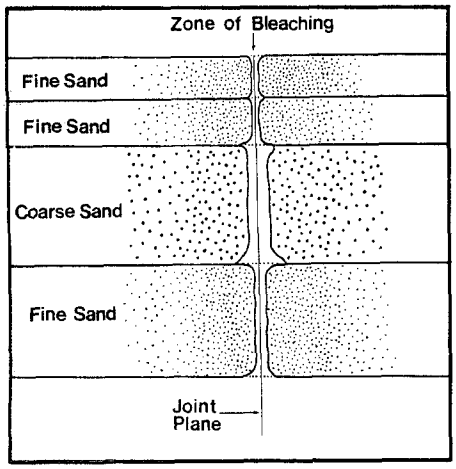


Figure 2. Sketch of the relationship of bleaching to jointing at Red Dome. Note that the extent of bleaching is related to formation porosity as determined by grain size (after Moulton, 1926).

authigenic in origin. A second reason that the study area is appropriate as a test of the hypothesis is that the timing of fracturing and oil migration is constrained by the geologic history. The dome, and probably the fractures and associated bleaching (Moulton, 1926), formed in the early Tertiary during the Laramide orogeny (Wilson, 1936). The precipitation of any authigenic magnetic minerals and acquisition of an associated remanence would therefore have to be younger than the fracturing. This constrains the time of acquisition of any remanence associated with the hydrocarbons.

BACKGROUND

Donovan and others (1979) reported magnetite in altered Permian rocks above the producing horizons at the Cement oil field in southern Oklahoma and suggested that hydrocarbons had migrated up along faults, causing reduction of hematite and iron hydro-oxides, and the precipitation of magnetite. Donovan and others (1979) also suggested that the authigenic magnetite found in the oil field could be the source of an aeromagnetic anomaly observed over the field. More recent studies of the Cement oil field reported that the magnetite contains synthetic textures and is probably related to drilling activity (Reynolds and others, 1985). Reynolds and others (1984, 1985) did report authigenic ferrimagnetic pyrrhotite in well cuttings from the Cement field and suggested that it precipitated as a result of hydrocarbon-related fluids.

Elmore and others (1987) and McCabe and others (1987) reported the occurrence of secondary magnetite in samples of solid bitumen. Geochemical analyses of the bitumens indicate that they had been extensively biodegraded. Elmore and others (1987) and McCabe and others (1987) suggested that the magnetite could have formed as a result of microbial activity during the diagenesis of the hydrocarbons.

A study of hydrocarbon-saturated speleothem cave deposits in southern Oklahoma also provides evidence of a relationship between hydrocarbons and a remanence residing in authigenic magnetite (Elmore and others, 1987). The speleothems, some of which contain hydrocarbons in primary fluid inclusions, are Permian in age, on the basis of interbedded fossils. Those speleothems which contain hydrocarbons have a Permian magnetization residing in authigenic magnetite which is more than an order of magnitude stronger than the magnetization in the speleothems which contain no hydrocarbons. These results suggest that the chemical conditions created by the hydrocarbons caused precipitation of authigenic magnetite and acquisition of the associated chemical remanent magnetization (CRM). Although the mechanism of the formation of the magnetite has not been determined, the evidence from the speleothems suggests that a genetic relationship between hydrocarbons and magnetite does exist.

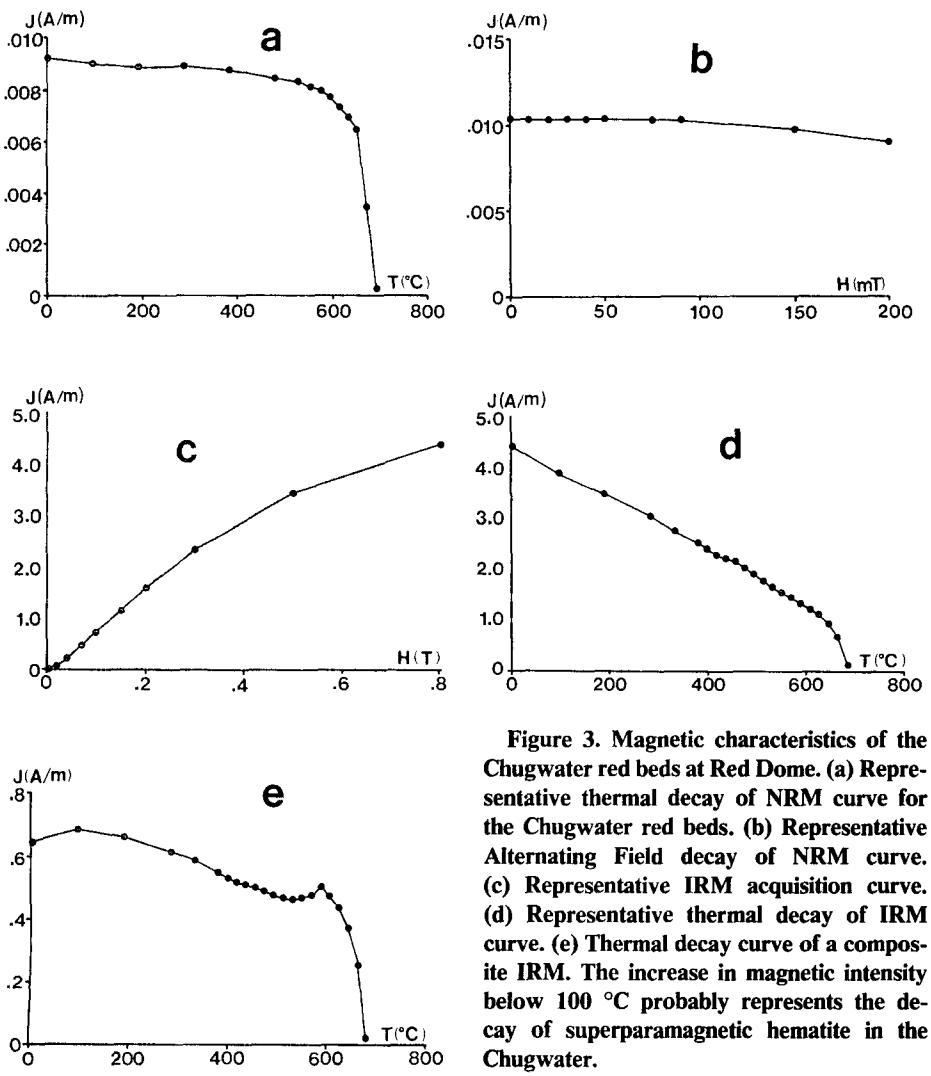


Figure 3. Magnetic characteristics of the Chugwater red beds at Red Dome. (a) Representative thermal decay of NRM curve for the Chugwater red beds. (b) Representative Alternating Field decay of NRM curve. (c) Representative IRM acquisition curve. (d) Representative thermal decay of IRM curve. (e) Thermal decay curve of a composite IRM. The increase in magnetic intensity below 100 °C probably represents the decay of superparamagnetic hematite in the Chugwater.

GEOLOGIC SETTING

Red Dome is a doubly plunging anticline, trending northwest to southeast, which exposes the upper portion of the Triassic Chugwater Formation. The Chugwater crops out as cliffs of interbedded thin siltstones, sandy shales, and sandstones (Wilson, 1938; Knappen and Moulton, 1930; Thom, 1922; Thom and others, 1935). Disseminated gypsum, rare gypsum beds, as well as gypsum veinlets, are common in the upper portion of the Chugwater. The shales, limestones, and calcareous sandstones of the Jurassic Sundance Formation unconformably overlie the Chugwater Formation. Locally, the Chugwater unconformably overlies the white

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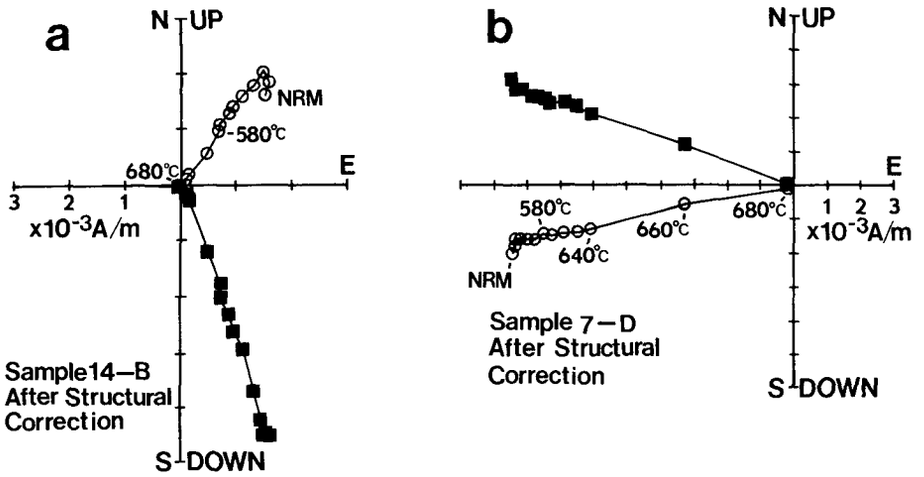


Figure 4. Orthogonal projections, illustrating demagnetization of representative specimens of the red beds. (a) Orthogonal projection of thermal decay of the NRM of the Chugwater red beds at Red Dome, displaying a south-southeasterly, and moderately upward direction. The squares represent the horizontal component of magnetization, and the circles represent the vertical component of magnetization. (b) Orthogonal projection of thermal decay of the NRM of the Chugwater Formation at Red Dome, displaying a west-northwesterly and slightly downdipping direction.

sandstones of the Pennsylvanian Tensleep Formation.

Red Dome is one of a series of domes, half domes, and anticlines which are aligned along a 175-km anticlinal trend called the "Nye-Bowler Lineament" (Wilson, 1936; Blackstone, 1940; Hoppin, 1974). The regional strike of the lineament is N65°W, but varies from N50°W to N87°W (Blackstone, 1940). The Lineament terminates abruptly to the east against a north-south fault, which forms the eastern boundary of the Pryor Mountains, and it terminates to the west near the northern boundary of the Beartooth uplift.

Stratigraphic studies of the area indicate that the dome formed in the Late Cretaceous or early Tertiary during the Laramide Orogeny (Alpha and Fanshawe, 1954; Wilson, 1936). The dome is probably a product of the surficial expression of left-lateral wrench faulting (David W. Stearns, 1987, personal commun.), associated with strike-slip faulting of basement rocks during the latter stages of Laramide deformation events in the Paleocene (Blackstone, 1940; Fanshawe, 1952; Thom, 1923, 1952; Wilson, 1936). Most of the crest of the dome, as well as the southwestern half of the dome, have been eroded away.

METHODS

Approximately 200 oriented-core samples were collected from 19 sample sites at Red Dome, using a portable drilling apparatus. Both core samples and hand samples were collected from the unaltered, bleached, and oil-saturated red beds. The natural remanent magnetization (NRM) in the specimens (2.2-cm length by 2.5-cm diameter) were measured, using a United Scientific three-axis cryogenic magnetometer in a magnetically shielded room where the magnetic field is less than 200 nanoTesla (nT). Stepwise alternating field or thermal demagnetization techniques were used to identify the components of magnetization. During AF demagnetization, the specimens were subjected to 11 steps up to field strengths of 200 milliTesla (mT) in a 2-axis tumbling unit. Thermal demagnetization was performed in a Schonstedt Thermal Specimen Demagnetizer (model TSD-1). Specimens were heated in a stepwise fashion between 100 and 700 °C (13 steps). During each heating step, specimens were held at a constant temperature for 30 minutes and then allowed to cool in a field of less than 2 nT. The demagnetization results were plotted on orthogonal projections (Zijderveld, 1967), and directions for components were calculated, using principle-component analysis (Kirschvink, 1980). Magnetic-susceptibility (MS) measurements were made with a Sapphire Instrument SI-2 system with a sensitivity of about 1×10^{-7} SI units.

Isothermal remanent magnetization (IRM) experiments were conducted to determine the coercivity spectra of the magnetic minerals. Specimens were subjected to increasing magnetic fields as great as 0.8 T at room temperature and measured after each step to determine the increase in magnetization.

A representative group of specimens was also given a composite IRM, using a technique similar to that used by Channell and others (1982). A composite IRM was produced by inducing an IRM of 0.80 T on a specimen, reversing the field direction 180°, and applying a 0.20-T field to the specimen. The 0.80-T field will remagnetize both the high- and low-coercivity minerals in the sample. The 0.20-T field, however, will remagnetize only the lower-coercivity phases present in the sample and in a direction directly opposite to that of the higher-coercivity phases. Thermal decay of a composite IRM will detect relatively small amounts of either low- or high-coercivity magnetic phases because the directions of the magnetization of the high- and low-coercivity phases are directly opposite to one another, which acts to amplify their differences.

After a specimen was given a composite or a regular IRM, the specimen was thermally demagnetized. The same demagnetization procedure was used as was used for the thermal demagnetization of NRM, except that several more demagnetization steps were added.

Petrographic studies were performed on all of the lithologies sampled. Thin sections were made for study under reflected light and with plain- and cross-polarized transmitted light. Small chips of the oil sands, the bleached red beds, and the unaltered red beds were examined with the scanning electron microscope to determine the detrital and authigenic phases present.

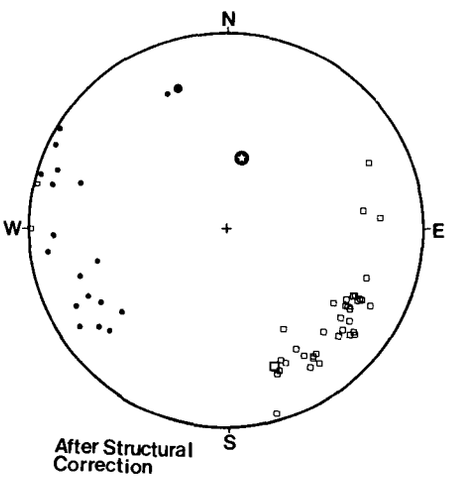


Figure 5. Schmidt equal-area projection of the after-structural correction magnetization directions for the Chugwater red beds at Red Dome. The dips are shallow at Red Dome, and the *in situ* directions (before structural correction) are similar to the tilt-corrected directions. Dots represent lower-hemisphere projections, and squares represent upper-hemisphere projections. The large dot and square represent the normal and reversed Triassic magnetic directions, respectively (Van der Voo and Grubbs, 1977). The star represents the modern magnetic direction.

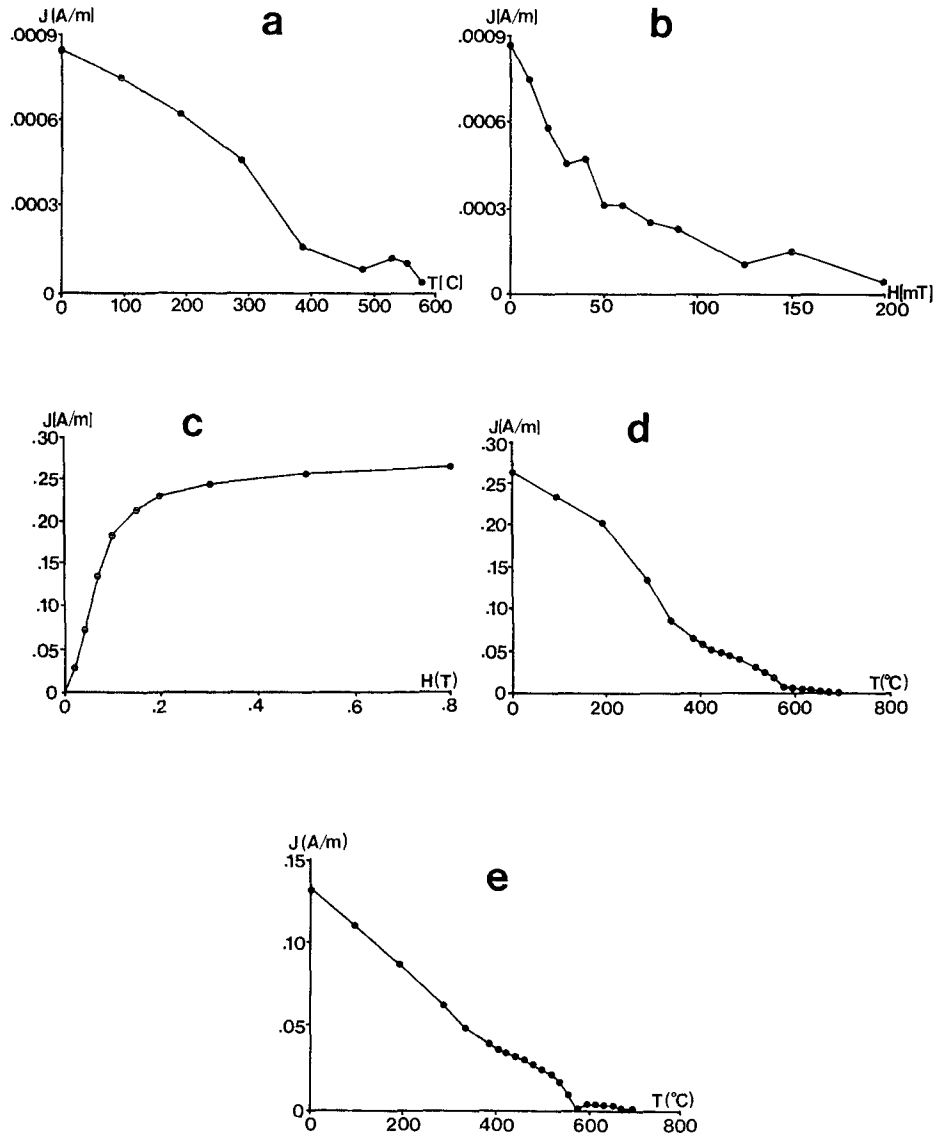


Figure 6. Magnetic characteristics of the oil-saturated sandstones. (a) Representative thermal decay of NRM curve for the oil-saturated sands. (b) Representative alternating field decay of NRM curve for the oil-saturated sands. (c) Representative IRM acquisition curve. (d) Representative thermal decay curve of IRM. (e) Thermal decay of composite IRM curve for the oil-saturated Chugwater Formation at Red Dome. The slight increase in magnetic intensity at ~580 °C represents the complete decay of magnetite. Decay up to 680 °C indicates the presence of a small quantity of hematite in the oil-saturated sandstones.

Samples of the three types of rocks were further studied by grinding them in a mortar and pestle and magnetically extracting the magnetic grains present. The magnetic extracts were studied by using scanning electron microscopy and energy-dispersive analysis (SEM-EDA).

RESULTS

Unaltered Red Beds

The unaltered red beds decay to 680 °C during thermal demagnetization and do not decay significantly during AF treatments up to 200 mT (Figs. 3a, 3b). These results, and the IRM acquisition and thermal decay experiments (Figs. 3c, 3d), indicate that high-coercivity hematite is the predominant magnetic mineral in the unaltered red beds.

Decay of composite IRM (Fig. 3e) suggests the presence of a small amount of magnetite in the unaltered red beds. The continuous decay in magnetic intensity as high as ~530 °C represents the decay of both the hematite and a small amount of magnetite in the sample. The increase in magnetic intensity beginning at ~530 °C presumably represents the rapid decay of magnetite as it approaches its Curie temperature. Since the magnetite is magnetized in a direction opposite to that of the hematite, its rapid rate of decay, when compared to the rate of decay of the hematite, causes a net increase in the magnetization of the sample. The maximum increase in magnetic intensity occurs at ~580 °C, the Curie point of magnetite, above which the remaining magnetization of the magnetite is destroyed. Above this temperature, the magnetism residing in the hematite decays until it is completely removed by 680 °C.

Paleomagnetic analysis of the red beds shows consistently stable linear decay with evidence of field reversals being recorded within the stratigraphic section sampled at Red Dome (Figs. 4a, 4b). The directions of magnetization in specimens from individual sites are consistent and show good statistical grouping (Table 1). The distribution of the magnetic directions from all of the sites, however, shows a counterclockwise streaked distribution away from the normal and reversed Triassic directions (Fig. 5). The data fit a great circle which passes through the normal and reversed Triassic directions. Although a fold test was not possible, previous studies of the Chugwater indicate that the unit contains a prefolding magnetization with inclinations similar to those observed in this study (for example, Shive and others, 1984; Herrero-Bervera and Helsley, 1983).

TABLE 1. SUMMARY OF PALEOMAGNETIC DATA FOR UNALTERED CHUGWATER RED-BED SAMPLE SITES AT RED DOME

	N/No.	*In situ* Decl./Incl.		After tilt correction		k	α_{95}	Paleopole after tilt correction	
		(°)	(°)	Decl./Incl.				Lat. (N)	Long. (E)
Site 2	5/6	144	-43	146	-18	77	9	44	120*
Site 3	6/6	116	-44	123	-28	240	4	34	148*
Site 4	10/10	114	-42	122	-26	179	4	32	148*
Site 7	7/7	281	23	289	7	57	8	16	150
Site 8	8/8	237	33	238	22	54	8	12	13
Site 11	6/6	132	-25	132	-21	236	4	37	137*
Site 14	3/3	161	-26	158	-26	623	5	54	108*
Site 15	6/6	263	5	264	16	37	11	2	172

Note: data shown for both before and after correction for bed dip. The value "k" is the precision parameter and indicates the relative dispersion of the magnetic directions (Fisher, 1953), and α_{95} represents the angular radius about the calculated site mean direction, within which there is a 95% probability that the true site mean direction lies. N/No. is the ratio of specimens analyzed to the number of specimens used in the statistical analysis.

*Denotes a reversed paleopole.

TABLE 2. SUMMARY OF PALEOMAGNETIC DATA FROM BLEACHED RED-BED SITES 9 AND 12 AT RED DOME FOR BEFORE AND AFTER STRUCTURAL CORRECTION

	N/No.	Decl./Incl.		k	α_{95}	Paleopole		X_1	X_2
		(°)	(°)			Lat.	Long.		
Before	17/18	341	67	346	2	76	171 W	3	3
After	17/18	334	62	251	2	72	164 E	3	4

Note: the value "k" is the precision parameter (Fisher, 1953), and the value α_{95} is the cone of 95% confidence. The values X_1 and X_2 represent the angular radius of the semiminor and semimajor axes in degrees, respectively, of the 95% elipse of confidence around the pole position (site 9, N216°E, 5°; site 12, N217°E, 9°).

Petrographic studies of the red beds indicate that they are well-sorted quartz sandstones. Black, well-rounded grains of specular hematite and secondary hematite pigment, which give the rocks their red color, are present.

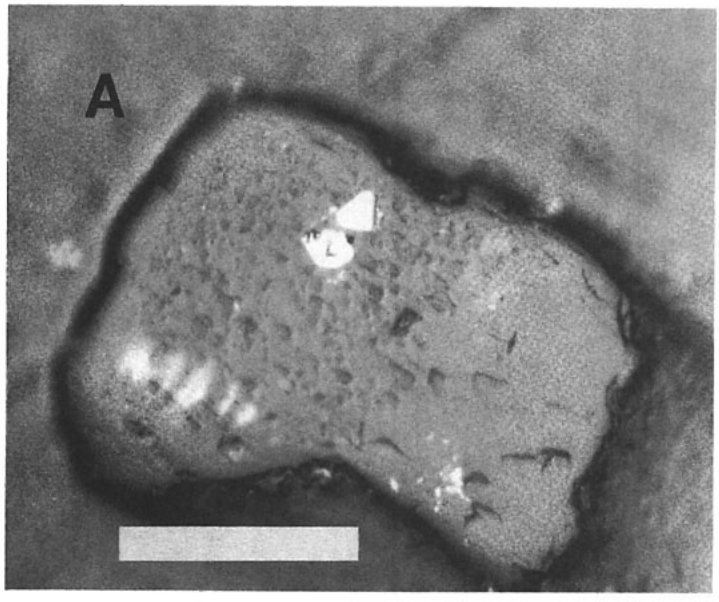
Oil-Saturated Red Beds

The magnetic characteristics of the oil-saturated sandstones are significantly different from those of the unaltered red beds. The NRM intensities of the oil-saturated sands are more than an order of magnitude less than those of the unaltered red beds (Fig. 6a) and decay up to 80 percent during AF treatments up to 200 mT (Fig. 6b). Although the oil-saturated sandstones possess a range of magnetic susceptibilities (5.1×10^{-5} – 4.0×10^{-4} SI units), the mean (1.0×10^{-4} SI; sd = 2.5×10^{-5}) is lower than the susceptibility for the red sandstones (2.5×10^{-4} SI; sd = 5×10^{-5}). The IRM acquisition and thermal decay experiments indicate that the magnetization is dominated by a low-coercivity phase with a maximum unblocking temperature of 580 °C (Figs. 6c, 6d). These results are consistent with magnetite as the predominant magnetic mineral in the oil-saturated sands. The presence of an inflection point in the IRM demagnetization curve at ~320 °C (Fig. 6d) suggests that pyrrhotite is also present in the samples. A gradual rise in the IRM acquisition curve above 0.30 T and minor decay above 580 °C during thermal decay of IRM (Figs. 6c, 6d) indicate that some hematite is present in these rocks.

Evidence from decay of composite IRM (Fig. 6e) also indicates the presence of magnetite, as well as a small amount of hematite, in the oil-saturated sandstones. The decay to zero intensity at 570 °C represents the point where the remaining antiparallel 0.20 T and 0.80 T IRMs are equal and cancel each other. The complete decay of magnetite is likely represented by the slight increase in the magnetic intensity between 570 and 590 °C. The fact that the peak in intensity is not exactly at 580 °C could be the result of the 20 °C measuring interval used during demagnetization and/or a slight miscalibration of the thermal demagnetizer. At higher temperatures, the magnetization resides only in hematite and shows continuous decay to 680 °C.

The oil-saturated sandstones generally display random, nonlinear decay patterns during both AF and thermal demagnetization. Due to the random magnetic directions obtained from the oil-saturated sands, paleomagnetic analysis was not possible.

The oil-saturated sandstones are composed of well-sorted, subangular quartz and are characterized by large pore spaces. Minor amounts of hematite pigment are observed, as well as small amounts of opaque minerals, including authi-



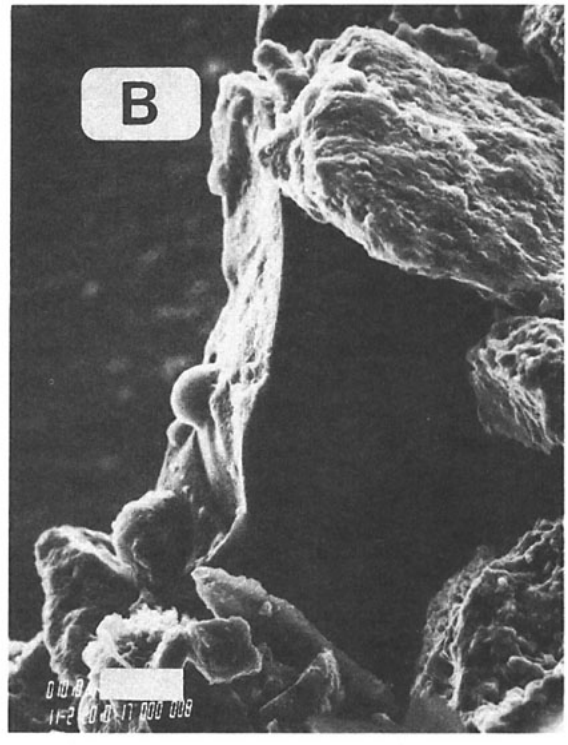


Figure 7. Magnetite in the oil-saturated sandstone. (A) Reflected-light photomicrograph of a quartz grain, which contains inclusions of magnetite. Scale bar is 100 microns. (B) Scanning electron microscope photomicrograph of a fragment of a spherical magnetic grain from the oil-saturated Chugwater Formation. Energy-dispersive analysis indicated that iron was the only detected element in this grain. Scale bar is 10 microns.

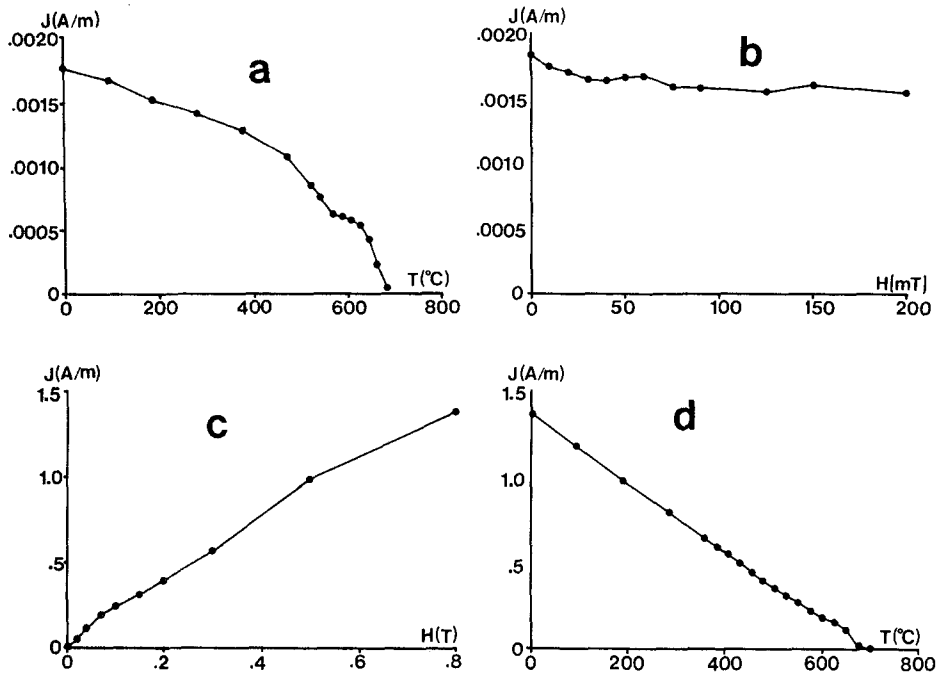


Figure 8. Magnetic characteristics of the bleached sandstones at Red Dome. (a) Representative thermal decay of NRM curve for the bleached sandstones. The abrupt drop in magnetic intensity at 580 °C and final decay at 680 °C indicates the presence of two magnetic phases with unblocking temperatures of 580 and 680 °C. (b) Alternating Field decay of NRM curve for the bleached sandstones. (c) Representative IRM acquisition curve for the bleached sandstones. (d) Thermal decay of IRM curve for the bleached Chugwater Formation at Red Dome.

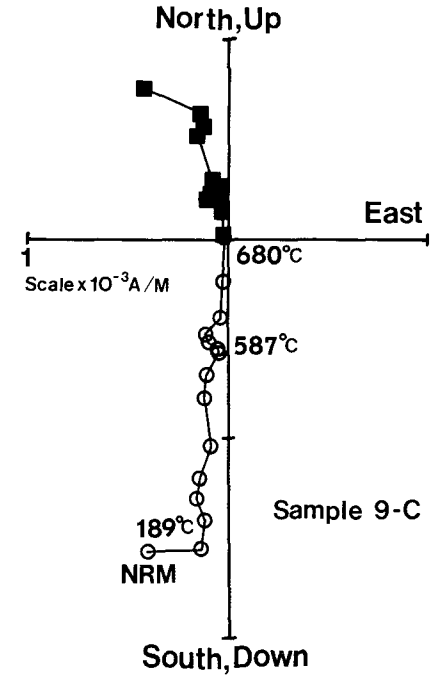


Figure 9. Orthogonal projection (before tilt correction) of thermal decay of NRM for a representative bleached specimen, displaying a north-northwesterly, steeply downdipping magnetic direction. The squares represent the horizontal component of magnetization, and the circles represent the vertical component of magnetization.

genic pyrite. Magnetite is present as inclusions within quartz grains within the magnetic extracts (Fig. 7A). Fragments of broken spheres have also been observed in the oil-saturated sand magnetic extracts (Fig. 7B). Energy dispersive analysis of the spheres indicated that iron was the only element detected in the fragments.

The Bleached Red Beds

The bleached specimens, which are not hydrocarbon saturated, generally show magnetic characteristics intermediate between the oil-saturated sands and the unaltered red beds. Thermal decay of the NRM suggests the presence of two magnetic phases with unblocking temperatures of ~580 and 680 °C (Fig. 8a). The magnetization does not decay significantly during AF demagnetization (Fig. 8b). The IRM acquisition and thermal decay experiments indicate that the magnetization is dominated by a high-coercivity mineral phase with a maximum unblocking temperature above 580 °C (Figs. 8c, 8d). These results are consistent with the high-coercivity mineral hematite being the predominant magnetic mineral in the bleached red beds. Composite IRM decay curves were inconsistent, and all that could be concluded from these experiments was that hematite was present in the bleached samples.

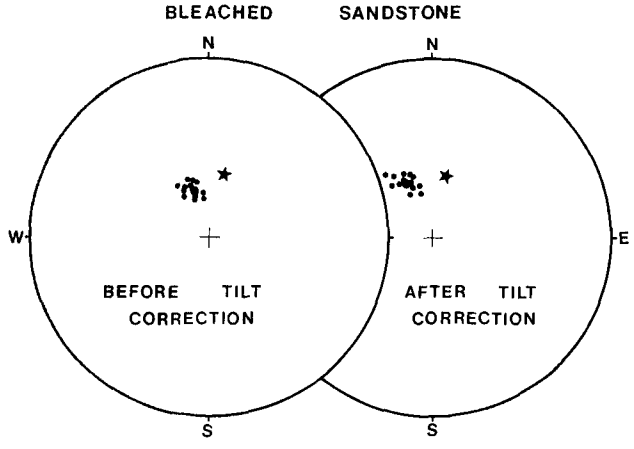


Figure 10. Schmidt equal-area projections for before and after tilt correction of the magnetic directions from the bleached red beds at Red Dome. Solid dots are lower-hemisphere projections. The large dots are the average magnetic directions of the data plotted. The radius of the large dot approximately corresponds to the radius of the α_{95} cone of confidence of the data. The star represents the modern magnetic direction.

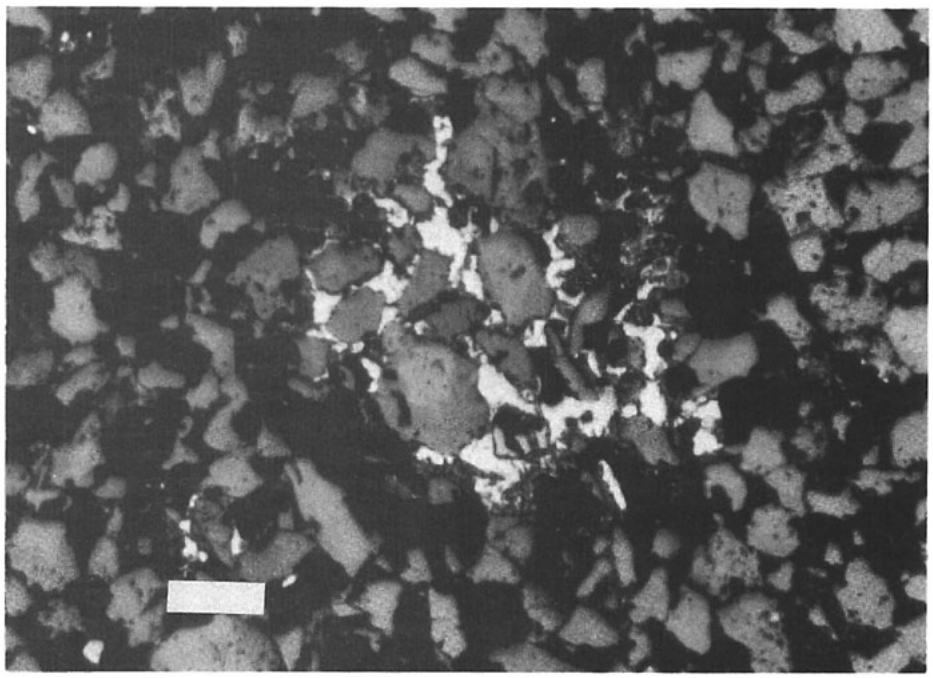


Figure 11. Photomicrograph of a sample from the bleached Chugwater Formation, shown under reflected light. The light gray/white concretion is authigenic specular hematite which has formed around the darker gray detrital quartz grains. This texture indicates that the specularite is secondary in origin. Scale bar is 100 microns.

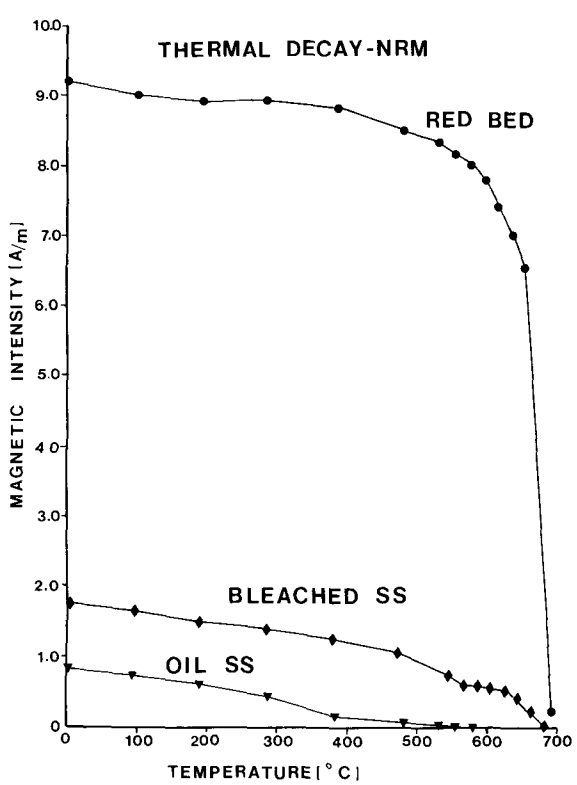


Figure 12. Summary figure shows the typical thermal decay curves of NRM for the three types of rocks sampled at Red Dome. Note the large difference in the magnetic intensity of the NRM between the red beds and the oil-saturated or the bleached sandstones. Intensity scale $\times 10^{-3}$ A/m.

Paleomagnetic analysis of the bleached red beds shows stable linear decay of NRM with consistent north-northwest and steeply downward-dipping directions (Fig. 9). The orientation of the rocks at the two sampling sites was only slightly different (Table 2), and limited exposures prevented sampling for a fold test. Statistical analysis of the data, however, did indicate that the magnetic directions cluster better before correction for bed dip than after correction for bed dip, although the difference is not significant at the 95 percent confidence level (Fig. 10 and Table 2).

Examination of the bleached sandstones in reflected light revealed the presence of abundant, authigenic specular hematite (Fig. 11). The authigenic hematite present in the bleached red beds differs markedly from that found in the unaltered red beds. The hematite found in the unaltered red beds occurs as discrete, well-rounded grains, which are detrital in origin, and as authigenic pigment. The specularite found in the bleached red beds displays a poikilitic texture that is clearly secondary.

DISCUSSION

The results of this study indicate that the three rock types at Red Dome have distinct magnetic characteristics. The differences in the NRM intensities of the red beds, bleached sandstones, and oil-impregnated sands are striking (Fig. 12). The NRM intensity of the red beds is approximately one order of magnitude greater than that measured in either the oil-saturated sands or the bleached sandstones. Although the difference is not as great, magnetic susceptibility is also higher in the red samples, compared to the oil-saturated sandstones.

In agreement with previous work by other investigators (for example, Shive and others, 1984; Collinson, 1974), the unaltered red beds contain a magnetization residing in hematite. Thermal decay of composite IRM indicates that a small quantity of magnetite is also present.

Paleomagnetic analysis of the red beds shows that the specimen directions obtained from individual sample sites are tightly clustered. Magnetic directions from all of the sites form a streaked, counterclockwise distribution, away from the normal and reversed Triassic directions. The distribution of these data could represent the vector addition of magnetizations acquired during reversals of the Earth's magnetic field in the Triassic. The consistently linear decay displayed by the magnetization in the red beds, however, suggests that this is unlikely.

An alternative interpretation is that the coun-

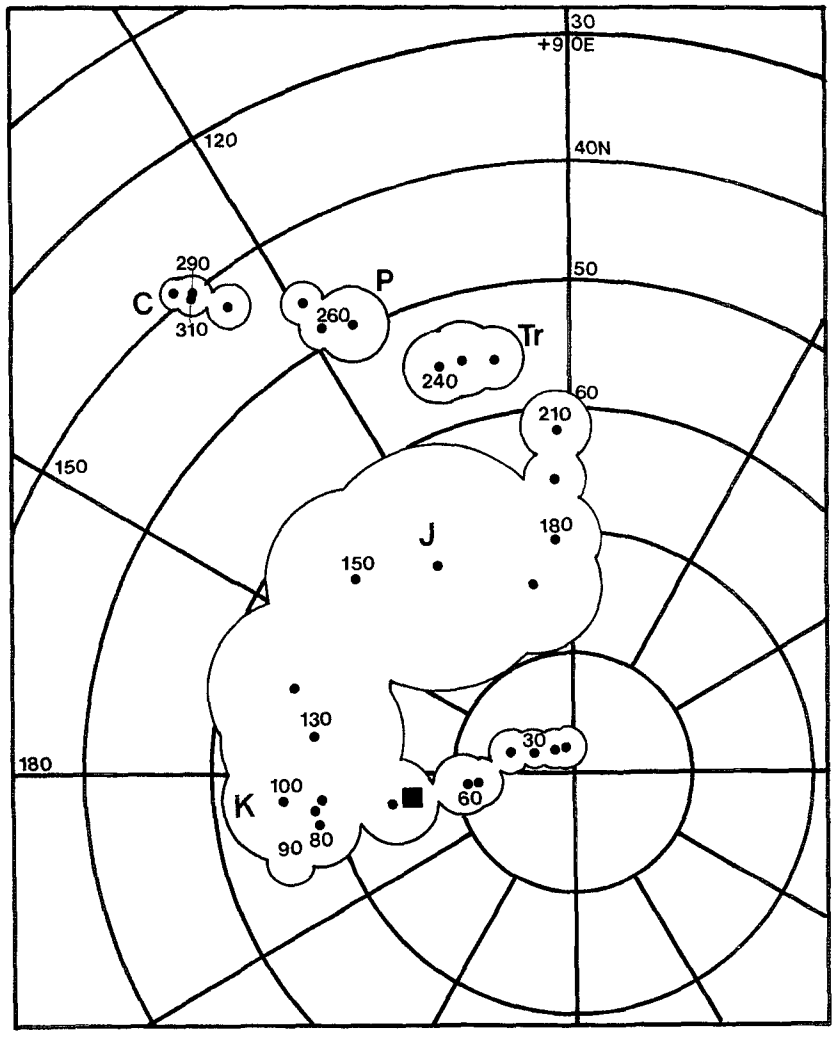


Figure 13. North American Apparent Polar Wander Path (after Irving and Irving, 1982); the paleopole (square) is determined from the magnetic direction obtained from the bleached sandstones at Red Dome.

terclockwise distribution of the magnetic directions could be the result of structural rotation associated with the formation of Red Dome. The orientation of the anticlinal ridge that forms Red Dome, and the types and orientation of faults found at Red Dome, suggest that Red Dome is a product of left-lateral wrench faulting. Because of this, it is possible that the red beds at Red Dome have been rotated counterclockwise from their original position. Because the magnetization of the red beds was acquired in the Triassic before Laramide deformation, rotation would produce a streaked distribution of the magnetic directions similar to what is observed at Red Dome today.

The magnetization in the oil-saturated sands is carried predominantly in magnetite. Petrographic and SEM-EDA studies of magnetic extracts show that both detrital and authigenic magnetite are present. The hydrocarbons caused the reduction, dissolution, and removal of hematite from the red beds. The removal of the hematite greatly reduced the magnetic intensity and resulted in the "uncovering" and relative enhancement of the magnetization carried by detrital magnetite. The chemical conditions created by the hydrocarbons apparently caused the precipitation of some authigenic magnetite and pyrrhotite. The dissolved iron from the red beds may have been the source of iron in the authigenic phases. Iron in oil (Ellrich and others, 1985) could also have been a source.

Although magnetic susceptibility is also reduced in the oil-impregnated samples, the relative decrease was not as great as for the NRM. This difference in relative decreases is presumably caused by the fact that the NRM is carried by the hematite, whereas the susceptibility is carried predominantly by magnetite. The oil-saturated sandstones lost most of their NRM and some susceptibility due to the dissolution of hematite. The detrital-magnetite and the authigenic-magnetite phases are responsible for the remaining susceptibility.

The magnetization in the bleached red beds is interpreted to be a CRM residing in authigenic hematite. Paleomagnetic analysis of the bleached red beds reveals the presence of magnetic directions which are consistently north-northwesterly and dipping steeply downward. Statistical analysis indicates that the directions cluster better before correction for bed dip than after correction for bed dip (Table 2), which suggests a post-folding magnetization. The paleopole determined for this magnetization, when plotted on the stable North American Apparent Polar Wander Path (Irving and Irving, 1982), indicates an early Tertiary age (Fig. 13). This time of remanence acquisition corresponds to the stratigraphically determined age of the dome (Wilson, 1936), which is the probable time of the formation of the fractures at the crest of the dome and the migration of hydrocarbons into the fractures. The hydrocarbons could not have migrated into Red Dome until the fractures formed. Therefore, the bleaching and the precipitation of the specularite could not have occurred until the early Tertiary, either during or after the formation of the dome.

The secondary hematite observed in the bleached samples occurs enclosing detrital quartz grains. This form of hematite is not observed in the unaltered Chugwater red beds (Collinson, 1974) and is not common in other well-studied red beds (Walker and others, 1981).

We suggest that the precipitation of authigenic specularite in the bleached red beds may have occurred as a secondary byproduct of the migration of hydrocarbons at Red Dome. As hydrocarbons migrated into the fractures at Red Dome, reducing conditions were also created in the sandstones adjacent to the fractures, possibly caused by the gases or fluids which are commonly associated with migrating hydrocarbons. This led to the dissolution of the hematite and bleaching in the red beds adjacent to the hydrocarbon-filled fractures. Subsequently, chemical conditions in the bleached sandstones changed, possibly caused by Laramide uplift and the erosion of the sediments overlying the Chugwater. This would have brought the

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Chugwater near the surface and closer to potentially oxidizing conditions. The oxidizing conditions led to the precipitation of authigenic hematite, presumably from iron-rich fluids in the bleached zones. The early Tertiary time of remanence acquisition in the bleached sandstones is consistent with this model.

The dissolution of hematite from the oil-impregnated and bleached sands probably resulted in the release of significant amounts of iron. A portion of this iron probably precipitated locally as magnetite, hematite, and iron sulfide. Some of the iron, however, may have been carried away by fluids. The dissolved hematite, therefore, could have been an important source of iron for diagenetic reactions in other units.

Donovan and others (1979) suggest that aeromagnetic surveys can be used as a relatively inexpensive hydrocarbon exploration tool. Previous work on hydrocarbon-saturated carbonate rocks (Benthien and Elmore, 1987; Elmore and others, 1987) indicates that the magnetic intensity of the oil-impregnated samples is approximately one order of magnitude greater than the magnetic intensity of the non–hydrocarbon-impregnated samples. The increase in magnetic intensity is believed to be caused by the precipitation of authigenic magnetite, which formed as a result of the chemical conditions created by the hydrocarbons. This suggests that looking for increases in the magnetic intensity of carbonate rocks, perhaps by means of aeromagnetic surveys, could be a viable hydrocarbon exploration tool. The results from the sandstones at Red Dome, however, indicate that the presence of hydrocarbons has resulted in a decrease in the NRM intensity by one order of magnitude and also a decrease in magnetic susceptibility. Although the hydrocarbons apparently caused some authigenic magnetite to precipitate, enough new magnetite did not form to make up for the intensity of magnetization and susceptibility lost when the hematite was removed. This suggests that searching for increases in magnetic intensity with aeromagnetic or other techniques may have questionable utility as a hydrocarbon exploration tool in sandstones similar to the Chugwater Formation.

It is apparent that hydrocarbons play an important role in several diagenetic reactions, some of which have paleomagnetic significance. Hydrocarbons can cause an increase in magnetization in some limestones but cause a decrease in red beds. Although other factors may also be important, the rock type is apparently one controlling element.

CONCLUSIONS

The results of this study have important implications for understanding hydrocarbon-related diagenesis in sandstones, understanding remagnetization mechanisms, and using aeromagnetic surveys as a hydrocarbon exploration tool. The presence of hydrocarbons in the red beds has caused the removal of hematite from the bleached and oil-impregnated sandstones, the "uncovering" of detrital magnetite and the precipitation of authigenic magnetite (and pyrrhotite) in the oil-saturated sandstones, and the precipitation of authigenic hematite in the bleached sandstones. By dissolving the hematite, the hydrocarbons caused a significant reduction in magnetic intensity and provided a source of iron for other diagenetic reactions.

Hydrocarbon-induced diagenesis did not remagnetize the rocks by acquisition of a stable secondary remanence residing in authigenic magnetite. The CRM in the bleached sandstones, however, is interpreted to be a byproduct of the hydrocarbon migration at Red Dome.

The results from Red Dome show that hydrocarbon migration into continental red beds produces effects that are very different from those previously observed in limestones. Depending on rock type, hydrocarbons can cause an increase or decrease in the magnetic intensity. This may limit the usefulness of aeromagnetic hydrocarbon exploration techniques in certain rock types (that is, red beds).

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